Please take five minutes to complete a course evaluation. You should get an e-mail with a link.

**Problem 1.** This problem will walk us through defining $\mathbb{Q}$ using an equivalence relation. Given $(a, b), (c, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, we say that

$$(a, b) \sim (c, d) \text{ if and only if } ad = bc.$$

For example $(1, 2) \sim (3, 6)$ but $(3, 4) \nsim (9, 4)$.

1. Show that $\sim$ is an equivalence relation.

2. Given $x \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, recall that **the equivalence class** of $x$ is

$$[x] = \{y \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\}) \mid x \sim y\}.$$

For example, $(-2, -6) \in [(1, 3)]$ but $(0, 4) \notin [(15, 25)]$. We can define

$$\mathbb{Q} = \{[x] \mid x \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})\}.$$

So $\mathbb{Q}$ is a set of equivalence classes. Why does it make sense to define $\mathbb{Q}$ in terms of equivalence classes (rather than just all pairs in $\mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$)?

3. Let $[(a, b)]$ and $[(c, d)]$ be equivalence classes as defined in Part (b). Given that we want these to represent rational numbers, tell me how we should define the following:

$$[(a, b)] + [(c, d)] = \ ?$$
$$[(a, b)] \cdot [(c, d)] = \ ?$$

Note: Each question mark should be replaced by some equivalence class. You do not have to formally prove your answer, but you should explain why it makes sense for $\mathbb{Q}$.

**Problem 2.** Assume that $A$ and $B$ are finite sets with $|A| = |B|$. Let $f\colon A \to B$ be a function. Prove that $f$ is injective if and only if $f$ is surjective.

**Problem 3.** Assume that $A$ and $B$ are finite sets with $|A| < |B|$. Prove that there is no surjective function $A \to B$.

*Note:* This is the "surjective analogue" of the Pigeonhole Principle that we did not prove in lecture.

*Hint:* Contrapositive. Assume $f\colon A \to B$ is surjective. Define $g\colon B \to A$ by choosing, for each $b \in B$ some $a_b \in f^{-1}(b)$. Show that $g$ is injective

**Problem 4.** Assume that a party has $n \geqslant 2$ attendees. Some people know each other and some people do not. Assuming that Person A knows Person B if and only if Person B knows Person A, show that there must be two people at the party who know the same number of people. (A person is not considered to know himself/herself so everyone knows between 0 people (if they know nobody at all) and $n - 1$ people (if they know everybody).)

*Hint*: Consider the case where there's some person who knows no one and then the case where everyone knows at least one person. Apply the Pigeonhole Principle to each.

**Problem 5.** Prove that the following sets are countably infinite. In other words, prove that they are equinumerous to $\mathbb{N}$.

1. The set of integers $F$ that are multiples of 5.

2. $A = \left\{\frac{1}{2^k} \mid k \in \mathbb{N}\right\}$.

3. $B = \{n \in \mathbb{Z} \mid n \geqslant -30\}$.